Day 1 - Recap

Inference Parameters ? → Fine-tune the model Response based on the use cases

① Temperature
- Low — very focused
- High — very creative

② Top p
- Low — 0.5
- High — 0.9

Cat 0.50
dog 0.30
hat 0.10

LLM → Transformer → Self Attention

③ Top K → 50
+
Top P → 0.50

Use case	Behavior	Temp	Top p	Top K	Max Tokens
① Customer Support / FAQ Bot	very deterministic, avoid hallucinations	0.2 - 0.3	0.7 - 0.8	50 - 60	200 - 300
② Content Summarizer	Concise, focused summaries, with minimal random-ness	0.2 - 0.4	0.8 - 0.9	50 - 100	250 - 300
③ Creative Storytelling	Highly creative, diverse, but less predictable.	0.7 - 1.0	0.9 - 1.0	100 - 200	300 - 500

Tips for Parameter Tuning

① Low temp + lower top p ⇒ Predictable, factual output
② High temp + High top p ⇒ Creative, varied responses
③ Max Tokens → for cost & control → Bills based on tokens.

RAG — Retrieval - Augmented Generation

① Take your OWN Document(s) → Repo → DB (Ext Knowledge Info)
② Find the MOST relevant part of a Q.
③ Send those relevant parts (Context) to LLM
④ Get an ANSWER grounded in Your Document

BIG PICTURE

PDF files (S3) → File extraction → chunking (Parsing) →
Model A embedding (tokens → vectors) → vector storage (open search DB) → search → send relevant chunks to model B → final ANSWER.

Langchain Components

① Models → connects LLMs
② Prompts → structure input
③ Memory → store conversation
④ Chains → workflow
⑤ Agents → decision logic
⑥ Tools → External APIs → calculator, web-search.

Problems:

- Context
- Memory
- Data retrieval
- Workflow
- Latency

LLMs → DoNOT remember conversations.

Demo: LANGCHAIN + Bedrock + DynamoDB

① Calling Model on Bedrock
② use Langchain Prompt Template & chains
③ Add (Chat History / Conversation History) — Dynamodb.

STAGE 1 — Send a msg to AI Model → get a reply
STAGE 2 — Structure the PROMPT
STAGE 3 → Store the previous messg

Amazon Bedrock Agents

↳ Dynamic Model.

System a req → reasons about it → decide what step to take next → use tools / KB → LLM → response.

Agentic System

Benefits of Bedrock Agents

① Managed Capabilities
② Choice of FMs / LLMs
③ Built-in tools
④ Memory Management
⑤ Multi-agent collaboration
⑥ Trace, debug, observability